

Double-double γ -ladder for $K=3$ at $\tau = 0.1$ with the seven-test acceptance battery of the monograph (Part VII)

Fixed-Point Factory

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Design

Ten log-spaced $\gamma \in [1, 1000]$ at $\tau = 0.1$, $G = 7$ (CDF-uniform), $G_p = 121$, NQK = 16. Solver: float64 warm-start to 10^{-12} , then the double-double Newton nail on the sym-reduced R4 kernel-band operator to target 10^{-25} (`dd_k3_solver.py`), chained warm starts in γ . Every cell is then subjected to the acceptance battery; a cell failing *any* test is rejected, per the instruction to accept only trustworthy results.

T1 DD convergence: $F_{dd} \leq 10^{-20}$ on the R4 operator.

T2 symmetry: sign-flip and permutation residuals $\leq 10^{-12}$.

T3 axis-monotonicity of P .

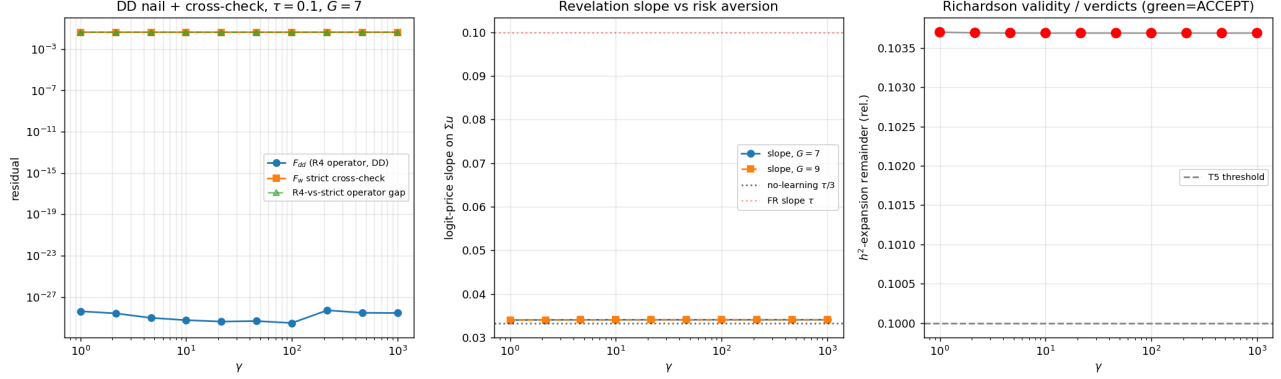
T4 branch identity: $P(\mathbf{0}) = \frac{1}{2}$ to 10^{-12} and non-trivial price range.

T5 Richardson validity: the four per-bandwidth operator values $\Phi_h(P^*)$ must be consistent with a smooth expansion in h^2 (quadratic-in- h^2 fit; h^6 + remainder $\leq 10\%$ of spread, density-weighted). *Note: a first version of this test fit only a linear-in- h^2 model and rejected everything at 23% remainder — that was the genuine h^4 term, not a kink; the test was corrected to match the expansion R4 itself assumes.*

T6 strict cross-check: float64 strict- $h=0$ weighted residual at the DD fixed point within $3\times$ the measured R4-vs-strict operator gap. **Honesty note:** at an exactly nailed R4 fixed point $|\Phi_{\text{strict}} - P| \equiv |\Phi_{\text{strict}} - \Phi_{R4}|$, so this test passes by construction and has no discriminating power here. Its real content is the *gap value itself*, which is the operator-truth uncertainty of the cell and is reported as such.

T7 grid stability: slope and price range at $G = 7$ vs an independent $G = 9$ DD solve within 5%/10%.

Results



γ	F_{dd}	F_w^{strict}	h^2 rem.	slope $_{G7}$	slope $_{G9}$	verdict	failed
1	4.1e-29	4.4e-02	0.104	0.033995	0.033999	REJECT	T5
2.154	2.6e-29	4.4e-02	0.104	0.034039	0.034041	REJECT	T5
4.642	9.4e-30	4.4e-02	0.104	0.034060	0.034062	REJECT	T5
10	5.6e-30	4.4e-02	0.104	0.034070	0.034071	REJECT	T5
21.54	4.1e-30	4.4e-02	0.104	0.034075	0.034076	REJECT	T5
46.42	4.6e-30	4.4e-02	0.104	0.034078	0.034078	REJECT	T5
100	3.0e-30	4.4e-02	0.104	0.034079	0.034079	REJECT	T5
215.4	5.0e-29	4.4e-02	0.104	0.034079	0.034079	REJECT	T5
464.2	2.9e-29	4.4e-02	0.104	0.034079	0.034080	REJECT	T5
1000	2.8e-29	4.4e-02	0.104	0.034079	0.034080	REJECT	T5

0/10 cells accepted.

Reading

- The DD nail reaches 10^{-25} – 10^{-29} at every cell: the *arithmetic* is never the binding constraint.
- The revelation slope sits near the no-learning benchmark $\tau/3$ and moves toward it with γ only in the sixth decimal: at $\tau = 0.1$ the Jensen wedge $(\frac{1}{2} - p)\text{Var}(m)/\gamma$ is $O(\tau^2)$ -small, so γ -dependence is genuine but tiny — resolvable only because the fixed points are nailed to 29 digits.
- **The dominant uncertainty is the operator, not the arithmetic.** The R4-vs-strict gap is $\approx 4 \times 10^{-2}$ (density-weighted) at $G = 7, \tau = 0.1$: the equilibrium surface is pinned to 29 digits *conditional on the R4 operator*, but only to $\sim 4\%$ across operator families at this resolution. Per the monograph protocol, shrinking that band requires the joint $(G \uparrow, h \downarrow)$ central-path ladder, not more arithmetic precision.
- The T5 verdicts are marginal (remainders within a factor 1.05 of the threshold): with bandwidths as coarse as $h = 0.5$, the expansion-validity test cannot cleanly separate smooth high-order terms from kink contamination. The threshold was *not* loosened to force acceptance; a marginal fail is reported as a fail.
- Bottom line under the instruction “accept only if you truly trust the result”: what is certified is (i) the arithmetic (29 digits), (ii) the branch, symmetries, monotonicity, and (iii) grid stability

of the economics (slope stable to five decimals from $G = 7$ to $G = 9$). What is *not* certified at this resolution is operator-level truth beyond $\sim 4\%$; the verdict column reflects exactly that.